# 2.4 Data annotation

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This handout exemplifies the workflow from a corpus query to an annotated dataset, covering the retrieval of concordance lines with quanteda, manual annotation in a spreadsheet, the handling of false hits, re-importing the data into R and extracting register information from the ICE text categories.

## Table of contents

Recommended reading . . . 1
Sample study . . . 1
Theoretical background . . . 1
Obtaining data . . . 2
Data annotation . . . 3
Dealing with problematic cases . . . 4
Getting the data back into R . . . 5
Adding a case list . . . 5
Adding text metadata . . . 6
Where do I go from here? . . . 10
References . . . 10

### Recommended reading

S. T. Gries (2013): Chapter 1.3.3

### Sample study

#### Theoretical background

Suppose we are interested in the object realisation patterns of the verb *eat* in the British ICE component. A quick review of the literature tells us that …

- … argument realisation may be related to the aspectual structure of a verbal action (cf. Goldberg 2001), but …
- … there seems to be a stronger emphasis on lexical aspect (telicity/atelicity; i.e., logical endpoints of actions) than on grammatical aspect (i.e., perfective/progressive).

Since grammatical aspect is also concerned with the temporal construal of actions, it raises the question of whether or not it can also influence argument realisation. To investigate the relationship between aspect and object realisation, we will perform an exemplary analysis on the verb lemma EAT.

### Obtaining data

We load all necessary libraries, query the ICE-GB corpus and run a KWIC search using the regular expression `^(eat(s|ing|en)?|ate)$`, which finds all inflectional forms of EAT (see 2.2 Regular expressions). We then store the results in a spreadsheet file `kwic_eat.xlsx`.

```
# Load libraries and corpus
library(quanteda)
library(writexl)
library(readxl)
library(tidyverse)

ICE_GB <- readRDS("ICE_GB.RDS")

# Perform query
kwic_eat <- kwic(ICE_GB,
                 "^(eat(s|ing|en)?|ate)$",
                 valuetype = "regex",
                 window = 15)

# Store results
write_xlsx(kwic_eat, "kwic_eat.xlsx")
```

When you open `kwic_eat.xlsx` in a spreadsheet software, the file will contain 7 columns by default (`docname`, `from`, `to`, `pre`, `keyword`, `post`, `pattern`). Each row corresponds to a match of your search expression in the corpus, which amounts to 110 here. This is your **raw output**.

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | **docname** | **from** | **to** | **pre** | **keyword** | **post** | **pattern** |
| 2 | ICE_GB/S1A-006.txt | 458 | 458 | had lunch < ICE-GB:S1A-006 #30 : 1 : A > Because I 'll have | eaten | anyway < ICE-GB:S1A-006 #31 : 1 : A > Well I mean part part of | \b(eat(s\|ing\|en)?\|ate)\b |
| 3 | ICE_GB/S1A-006.txt | 478 | 478 | : 1 : A > Well I mean part part of the reason I am | eating | will be so that I we don't have to have a picnic < ICE-GB:S1A-006 #32 | \b(eat(s\|ing\|en)?\|ate)\b |
| 4 | ICE_GB/S1A-006.txt | 785 | 785 | > Yeah < ICE-GB:S1A-006 #50 : 1 : A > So I 'd rather | eat | beforehand just to avoid uh <, , > any problems there < ICE-GB:S1A-006 #51 | \b(eat(s\|ing\|en)?\|ate)\b |
| 5 | ICE_GB/S1A-009.txt | 1198 | 1198 | the summer < ICE-GB:S1A-009 #70 : 1 : A > I must <, > | eat | them < ICE-GB:S1A-009 #71 : 1 : A > Yes < ICE-GB:S1A-009 #72 : 1 | \b(eat(s\|ing\|en)?\|ate)\b |
| 6 | ICE_GB/S1A-009.txt | 4529 | 4529 | > Yeah <, > < ICE-GB:S1A-009 #289 : 1 : B > We actually | ate | in their place < ICE-GB:S1A-009 #290 : 1 : B > A fish platter < | \b(eat(s\|ing\|en)?\|ate)\b |
| 7 | ICE_GB/S1A-010.txt | 958 | 958 | know I mean it would seem to be squandering it to <, > actually | eat | it for one 's own enjoyment <, , > < ICE-GB:S1A-010 #49 : | \b(eat(s\|ing\|en)?\|ate)\b |
| 8 | ICE_GB/S1A-010.txt | 2741 | 2741 | A > Yeah < ICE-GB:S1A-010 #168 : 1 : A > I 'm busy | eating | as a matter of fact <, > < ICE-GB:S1A-010 #169 : 1 : B | \b(eat(s\|ing\|en)?\|ate)\b |
| 9 | ICE_GB/S1A-011.txt | 3215 | 3215 | had a succession of them < ICE-GB:S1A-011 #197 : 2 : A > I have | eaten | eaten eaten < ICE-GB:S1A-011 #198 : 2 : A > I haven't been able to | \b(eat(s\|ing\|en)?\|ate)\b |
| 10 | ICE_GB/S1A-011.txt | 3216 | 3216 | a succession of them < ICE-GB:S1A-011 #197 : 2 : A > I have eaten | eaten | eaten < ICE-GB:S1A-011 #198 : 2 : A > I haven't been able to stop | \b(eat(s\|ing\|en)?\|ate)\b |
| 11 | ICE_GB/S1A-011.txt | 3217 | 3217 | succession of them < ICE-GB:S1A-011 #197 : 2 : A > I have eaten eaten | eaten | < ICE-GB:S1A-011 #198 : 2 : A > I haven't been able to stop < | \b(eat(s\|ing\|en)?\|ate)\b |
| 12 | ICE_GB/S1A-011.txt | 3245 | 3245 | to stop <, > < ICE-GB:S1A-011 #199 : 2 : A > I have | eaten | my way round the Yorkshire Dales < ICE-GB:S1A-011 #200 : 2 : B > Do | \b(eat(s\|ing\|en)?\|ate)\b |
| 13 | ICE_GB/S1A-011.txt | 4159 | 4159 | < ICE-GB:S1A-011 #249 : 2 : B > And so I ended up uhm just | eating | sort of lumps of chicken and things < ICE-GB:S1A-011 #250 : 2 : B > | \b(eat(s\|ing\|en)?\|ate)\b |
| 14 | ICE_GB/S1A-018.txt | 455 | 455 | : 1 : A > What you should do is order one first and then | eat | it and then sort of carry on from there < laughter > <, > | \b(eat(s\|ing\|en)?\|ate)\b |
| 15 | ICE_GB/S1A-018.txt | 498 | 498 | > <, > < ICE-GB:S1A-018 #28 : 1 : A > The bargain hunting | eat | < ICE-GB:S1A-018 #29 : 1 : B > So all right what did I have | \b(eat(s\|ing\|en)?\|ate)\b |
| 16 | ICE_GB/S1A-019.txt | 1038 | 1038 | food all laid out < ICE-GB:S1A-019 #75 : 1 : A > and everybody was | eating | something < ICE-GB:S1A-019 #76 : 1 : A > And it was a beautiful day | \b(eat(s\|ing\|en)?\|ate)\b |
| 17 | ICE_GB/S1A-022.txt | 3739 | 3739 | <, , > < ICE-GB:S1A-022 #247 : 1 : C > I 'm | eating | Helen 's with that < ICE-GB:S1A-022 #248 : 1 : D > But I | \b(eat(s\|ing\|en)?\|ate)\b |
| 18 | ICE_GB/S1A-023.txt | 1763 | 1763 | : B > Well he 's not really but he ha he just hasn't | eaten | anything for about two weeks now <, , > < ICE-GB:S1A-023 #116 : 1 | \b(eat(s\|ing\|en)?\|ate)\b |
| 19 | ICE_GB/S1A-023.txt | 1853 | 1853 | <, , > < ICE-GB:S1A-023 #121 : 1 : B > Oh name please | eat | something <, , > < ICE-GB:S1A-023 #122 : 1 : A > Oh actually | \b(eat(s\|ing\|en)?\|ate)\b |
| 20 | ICE_GB/S1A-023.txt | 2511 | 2511 | it seems lighter < ICE-GB:S1A-023 #165 : 1 : A > Well has some been | eaten | or uh < ICE-GB:S1A-023 #166 : 1 : B > Yeah I know that < | \b(eat(s\|ing\|en)?\|ate)\b |
| 21 | ICE_GB/S1A-025.txt | 5044 | 5044 | ICE-GB:S1A-025 #315 : 1 : A > They they they if they become hungry the | eat | the book <, , > and read the uh <, > and read | \b(eat(s\|ing\|en)?\|ate)\b |
| 22 | ICE_GB/S1A-028.txt | 2792 | 2792 | Sorry I 'm a bit vague < ICE-GB:S1A-028 #174 : 1 : B > | Eating | them before < unclear-words > < ICE-GB:S1A-028 #175 : 1 : A > Those are | \b(eat(s\|ing\|en)?\|ate)\b |
| 23 | ICE_GB/S1A-038.txt | 938 | 938 | < ICE-GB:S1A-038 #58 : 1 : C > She wouldn't go to sleep she wouldn't | eat | she wouldn't uch she wouldn't do anything <, > < ICE-GB:S1A-038 #59 : 1 | \b(eat(s\|ing\|en)?\|ate)\b |
| 24 | ICE_GB/S1A-046.txt | 5392 | 5392 | < ICE-GB:S1A-046 #414 : 1 : A > Do you think you would like to | eat | < ICE-GB:S1A-046 #415 : 1 : A > Would anybody like to eat I mean | \b(eat(s\|ing\|en)?\|ate)\b |
| 25 | ICE_GB/S1A-046.txt | 5405 | 5405 | like to eat < ICE-GB:S1A-046 #415 : 1 : A > Would anybody like to | eat | I mean < unclear-words > for breakfast < ICE-GB:S1A-046 #416 : 1 : A > | \b(eat(s\|ing\|en)?\|ate)\b |
| 26 | ICE_GB/S1A-047.txt | 2174 | 2174 | Z > < ICE-GB:S1A-047 #139 : 1 : C > Do you want anything to | eat | < ICE-GB:S1A-047 #140 : 1 : B > He 's going to say he | \b(eat(s\|ing\|en)?\|ate)\b |
| 27 | ICE_GB/S1A-047.txt | 2573 | 2573 | , > < ICE-GB:S1A-047 #163 : 1 : C > Do you want anything to | eat | < ICE-GB:S1A-047 #164 : 1 : B > No thanks < ICE-GB:S1A-047 #165 : 1 | \b(eat(s\|ing\|en)?\|ate)\b |
| 28 | ICE_GB/S1A-047.txt | 2629 | 2629 | > No < ICE-GB:S1A-047 #168 : 1 : C > Do you want anything to | eat | < ICE-GB:S1A-047 #X169 : 1 : Z > < ICE-GB:S1A-047 #170 : 1 : C | \b(eat(s\|ing\|en)?\|ate)\b |
| 29 | ICE_GB/S1A-051.txt | 473 | 473 | you think <, > < ICE-GB:S1A-051 #32 : 1 : B > Well I | eat | one good cooked meal a day of meat and vegetables <, , > < | \b(eat(s\|ing\|en)?\|ate)\b |
| 30 | ICE_GB/S1A-051.txt | 620 | 620 | ICE-GB:S1A-051 #41 : 1 : B > I don't always feel I 'm like | eating | it or wanting it but I do it < ICE-GB:S1A-051 #42 : 1 : A | \b(eat(s\|ing\|en)?\|ate)\b |
| 31 | ICE_GB/S1A-055.txt | 1388 | 1388 | : C > The bloke in charge always he knows Ben quite well cos he | eats | there a lot <, > < ICE-GB:S1A-055 #95 : 1 : C > he | \b(eat(s\|ing\|en)?\|ate)\b |
| 32 | ICE_GB/S1A-055.txt | 1848 | 1848 | a video and <, > suppose I have to cook something otherwise I just | eat | crisps all weekend < ICE-GB:S1A-055 #121 : 1 : C > What did you do | \b(eat(s\|ing\|en)?\|ate)\b |
| 33 | ICE_GB/S1A-056.txt | 1755 | 1755 | They come from trees < ICE-GB:S1A-056 #107 : 2 : B > Human beings ca | eat | them < ICE-GB:S1A-056 #X108 : 2 : Z > < ICE-GB:S1A-056 #X109 : 2 : | \b(eat(s\|ing\|en)?\|ate)\b |
| 34 | ICE_GB/S1A-059.txt | 880 | 880 | <, , > I still find that uh <, , > when I | eat | I haven't <, , > I haven't been ab able < ICE-GB:S1A-059 #47 : | \b(eat(s\|ing\|en)?\|ate)\b |
| 35 | ICE_GB/S1A-059.txt | 930 | 930 | that I <, , > probably I know that I should <, > | eat | but <, , > when and I cook <, > uh considerable < | \b(eat(s\|ing\|en)?\|ate)\b |
| 36 | ICE_GB/S1A-059.txt | 1280 | 1280 | , , > < ICE-GB:S1A-059 #61 : 1 : A > But you worry about | eating | < ICE-GB:S1A-059 #62 : 1 : A > Is that is that something new < | \b(eat(s\|ing\|en)?\|ate)\b |
| 37 | ICE_GB/S1A-059.txt | 1665 | 1665 | <, > are you the sort of cook who < unclear-words > forward to | eating | his own food < ICE-GB:S1A-059 #84 : 1 : B > < unclear-words > < | \b(eat(s\|ing\|en)?\|ate)\b |
| 38 | ICE_GB/S1A-059.txt | 2230 | 2230 | > Mhm <, , > < ICE-GB:S1A-059 #120 : 1 : B > I | eat | a <, > very balanced diet < ICE-GB:S1A-059 #121 : 1 : B > | \b(eat(s\|ing\|en)?\|ate)\b |
| 39 | ICE_GB/S1A-059.txt | 2578 | 2578 | : A > Mm < ICE-GB:S1A-059 #140 : 1 : A > D 'you | eat | on your own < ICE-GB:S1A-059 #141 : 1 : B > Yes <, , | \b(eat(s\|ing\|en)?\|ate)\b |

> **Why not use word boundaries?**
>
> Word boundaries `\\b` are often used to delimit words in a regular expression. Since `kwic()` matches each token separately, however, `\\b` also allows matches within hyphenated tokens: The pattern `\\b(eat(s|ing|en)?|ate)\\b` finds *eating* in the compounds *egg-eating* and *bone-eating*, too. The anchors `^` and `$`, by contrast, require the entire token to match.

## Data annotation

Whenever you decide to work on your corpus results, it is good practice to duplicate your file and append the current date to the filename. Re-save it as, for instance, `kwic_eat_2024-09-09.xlsx` and open it again. This way you are performing basic **version control**, which will allow you to return to previous stages of your analysis with ease. Writing the date in the format year-month-day ensures that your files are also sorted chronologically in your file browser.

In your spreadsheet software, you can now add your variables of interest to the empty columns next to your output data. For our specific example, we will need one that captures object realisation and one the type of verb aspect. Let's simply call them `object_realisation` and `verb_aspect`.

> **Naming variables**
>
> Of course, you could also opt for a different column name, as long as it has **no spaces** or **special characters** (e.g., !?%#). You could name the first column `object_realisation` or, even more plainly, `object`, but **not** `direct object` or `object realisation` with spaces. Otherwise you are bound to encounter a surge of cryptic error messages in your R console.

Now, you are ready to annotate your data! An easy coding scheme would involve classifying rows where *eat* occurs with an object as `yes`. Conversely, rows where the direct object is not realised syntactically are assigned the column value `no`. In the aspect column, verbal aspect will be coded as either `perfective`, `progressive` or `neutral`, following Gries (2014: 118).

| | pre | keyword | post | pattern | object_realisation | aspect_verb |
|---|---|---|---|---|---|---|
| 2 | had lunch < ICE-GB:S1A-006 #30 : 1 : A > Because I ' ll have | eaten | anyway < ICE-GB:S1A-006 #31 : 1 : A > Well I mean part part of | \b(eat(s\|ing\|en)?\|ate)\b | no | perfective |
| 3 | : 1 : A > Well I mean part part of the reason I am | eating | will be so that I we don't have to have a picnic < ICE-GB:S1A-006 #32 | \b(eat(s\|ing\|en)?\|ate)\b | no | progressive |
| 4 | > Yeah < ICE-GB:S1A-006 #50 : 1 : A > So I ' d rather | eat | beforehand just to avoid uh < , , > any problems there < ICE-GB:S1A-006 #51 | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 5 | the summer < ICE-GB:S1A-009 #70 : 1 : A > I must < , > | eat | them < ICE-GB:S1A-009 #71 : 1 : A > Yes < ICE-GB:S1A-009 #72 : 1 | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 6 | > Yeah < , > < ICE-GB:S1A-009 #289 : 1 : B > We actually | ate | in their place < ICE-GB:S1A-009 #290 : 1 : B > A fish platter < | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 7 | know I mean it would seem to be squandering it to < , > actually | eat | it for one ' s own enjoyment < , , > < ICE-GB:S1A-010 #49 : | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 8 | A > Yeah < ICE-GB:S1A-010 #168 : 1 : A > I ' m busy | eating | as a matter of fact < , > < ICE-GB:S1A-010 #169 : 1 : B | \b(eat(s\|ing\|en)?\|ate)\b | no | progressive |
| 9 | had a succession of them < ICE-GB:S1A-011 #197 : 2 : A > I have | eaten | eaten eaten < ICE-GB:S1A-011 #198 : 2 : A > I haven't been able to | \b(eat(s\|ing\|en)?\|ate)\b | no | perfective |
| 10 | a succession of them < ICE-GB:S1A-011 #197 : 2 : A > I have eaten | eaten | eaten < ICE-GB:S1A-011 #198 : 2 : A > I haven't been able to stop | \b(eat(s\|ing\|en)?\|ate)\b | no | perfective |
| 11 | succession of them < ICE-GB:S1A-011 #197 : 2 : A > I have eaten eaten | eaten | < ICE-GB:S1A-011 #198 : 2 : A > I haven't been able to stop < | \b(eat(s\|ing\|en)?\|ate)\b | no | perfective |
| 12 | to stop < , > < ICE-GB:S1A-011 #199 : 2 : A > I have | eaten | my way round the Yorkshire Dales < ICE-GB:S1A-011 #200 : 2 : B > Do | \b(eat(s\|ing\|en)?\|ate)\b | no | perfective |
| 13 | < ICE-GB:S1A-011 #249 : 2 : B > And so I ended up uhm just | eating | sort of lumps of chicken and things < ICE-GB:S1A-011 #250 : 2 : B > | \b(eat(s\|ing\|en)?\|ate)\b | yes | progressive |
| 14 | : 1 : A > What you should do is order one first and then | eat | it and then sort of carry on from there < laughter > < , > | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 15 | > < , > < ICE-GB:S1A-018 #28 : 1 : A > The bargain hunting | eat | < ICE-GB:S1A-018 #29 : 1 : B > So all right what did I have | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 16 | food all laid out < ICE-GB:S1A-019 #75 : 1 : A > and everybody was | eating | something < ICE-GB:S1A-019 #76 : 1 : A > And it was a beautiful day | \b(eat(s\|ing\|en)?\|ate)\b | yes | progressive |
| 17 | < , , > < ICE-GB:S1A-022 #247 : 1 : C > I ' m | eating | Helen ' s with that < ICE-GB:S1A-022 #248 : 1 : D > But I | \b(eat(s\|ing\|en)?\|ate)\b | yes | progressive |
| 18 | : B > Well he ' s not really but he ha he just hasn't | eaten | anything for about two weeks now < , , > < ICE-GB:S1A-023 #116 : 1 | \b(eat(s\|ing\|en)?\|ate)\b | yes | perfective |
| 19 | < , , > < ICE-GB:S1A-023 #121 : 1 : B > Oh name please | eat | something < , , > < ICE-GB:S1A-023 #122 : 1 : A > Oh actually | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 20 | it seems lighter < ICE-GB:S1A-023 #165 : 1 : A > Well has some been | eaten | or uh < ICE-GB:S1A-023 #166 : 1 : B > Yeah I know that < | \b(eat(s\|ing\|en)?\|ate)\b | no | perfective |
| 21 | ICE-GB:S1A-025 #315 : 1 : A > They they they if they become hungry they | eat | the book < , , > and read the uh < , > and read | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 22 | Sorry I ' m a bit vague < ICE-GB:S1A-028 #174 : 1 : B > | Eating | them before < unclear-words > < ICE-GB:S1A-028 #175 : 1 : A > Those are | \b(eat(s\|ing\|en)?\|ate)\b | yes | progressive |
| 23 | < ICE-GB:S1A-038 #58 : 1 : C > She wouldn't go to sleep she wouldn't | eat | she wouldn't uch she wouldn't do anything < , > < ICE-GB:S1A-038 #59 : 1 | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 24 | < ICE-GB:S1A-046 #414 : 1 : A > Do you think you would like to | eat | < ICE-GB:S1A-046 #415 : 1 : A > Would anybody like to eat I mean | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 25 | like to eat < ICE-GB:S1A-046 #415 : 1 : A > Would anybody like to | eat | I mean < unclear-words > for breakfast < ICE-GB:S1A-046 #416 : 1 : A > | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 26 | Z > < ICE-GB:S1A-047 #139 : 1 : C > Do you want anything to | eat | < ICE-GB:S1A-047 #140 : 1 : B > He ' s going to say he | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 27 | , > < ICE-GB:S1A-047 #163 : 1 : C > Do you want anything to | eat | < ICE-GB:S1A-047 #164 : 1 : B > No thanks < ICE-GB:S1A-047 #165 : 1 | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 28 | > No < ICE-GB:S1A-047 #168 : 1 : C > Do you want anything to | eat | < ICE-GB:S1A-047 #X169 : 1 : Z > < ICE-GB:S1A-047 #170 : 1 : C | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 29 | you think < , > < ICE-GB:S1A-051 #32 : 1 : B > Well I | eat | one good cooked meal a day of meat and vegetables < , > < | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 30 | ICE-GB:S1A-051 #41 : 1 : B > I don't always feel I ' m like | eating | it or wanting it but I do it < ICE-GB:S1A-051 #42 : 1 : A | \b(eat(s\|ing\|en)?\|ate)\b | yes | progressive |
| 31 | : C > The bloke in charge always he knows Ben quite well cos he | eats | there a lot < , > < ICE-GB:S1A-055 #95 : 1 : C > he | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 32 | a video and < , > suppose I have to cook something otherwise I just | eat | crisps all weekend < ICE-GB:S1A-055 #121 : 1 : C > What did you do | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 33 | They come from trees < ICE-GB:S1A-056 #107 : 2 : B > Human beings can | eat | them < ICE-GB:S1A-056 #X108 : 2 : Z > < ICE-GB:S1A-056 #X109 : 2 : | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |
| 34 | < , , > I still find that uh < , , > when I | eat | I haven't < , , > I haven't been ab able < ICE-GB:S1A-059 #47 : | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 35 | that I < , , > probably I know that I should < , > | eat | but < , , > when and I cook < , > uh considerable < | \b(eat(s\|ing\|en)?\|ate)\b | no | neutral |
| 36 | , , > < ICE-GB:S1A-059 #61 : 1 : A > But you worry about | eating | < ICE-GB:S1A-059 #62 : 1 : A > Is that is that something new < | \b(eat(s\|ing\|en)?\|ate)\b | no | progressive |
| 37 | < , > are you the sort of cook who < unclear-words > forward to | eating | his own food < ICE-GB:S1A-059 #84 : 1 : B > < unclear-words > < | \b(eat(s\|ing\|en)?\|ate)\b | yes | progressive |
| 38 | > Mhm < , , > < ICE-GB:S1A-059 #120 : 1 : B > I | eat | a < , > very balanced diet < ICE-GB:S1A-059 #121 : 1 : B > | \b(eat(s\|ing\|en)?\|ate)\b | yes | neutral |

### Dealing with problematic cases

However, things are not always that clear-cut. What if you encounter a false positive, i.e., an erroneous hit in your dataset? In the following concordance line, the keyword *ate* is actually part of the preceding word, *inappropriate*, which has been split across two lines in the corpus file.

| docname | pre | keyword | post |
|---|---|---|---|
| ICE_GB/W2F-019.txt | -> Too much colour on her face would be inappropri < l > | ate | , she feels , but she wears a light foundation . < ICE-GB:W2F-019 #28 : |

> ❗ What do I do with false hits?
>
> Short answer: Do **not** delete irrelevant rows or columns. Essentially, from the moment you've obtained your corpus output, you should withstand the temptation to delete anything from it. Instead, adopt the practice of indicating missing values or irrelevant rows by an `NA` in a separate column. In later analyses, these can be easily filtered out! This also minimises the risk of accidentally getting rid of data that could have proven important at a later point in time.

### Getting the data back into R

Import the Excel file via

```
# Read file contents into the variable "kwic_data"
kwic_data <- read_xlsx("kwic_eat_2024-09-09.xlsx")
```

```
# Print the first six lines of "kwic_data"
print(head(kwic_data))
```

```
# A tibble: 6 x 9
  docname  from    to pre   keyword post  pattern object_realisation verb_aspect
  <chr>   <dbl> <dbl> <chr> <chr>   <chr> <chr>   <chr>              <chr>
1 ICE_GB~   458   458 had ~ eaten   anyw~ "\\b(e~ no                 perfective
2 ICE_GB~   478   478 : 1 ~ eating  will~ "\\b(e~ no                 progressive
3 ICE_GB~   785   785 > Ye~ eat     befo~ "\\b(e~ no                 neutral
4 ICE_GB~  1198  1198 the ~ eat     them~ "\\b(e~ yes                neutral
5 ICE_GB~  4529  4529 > Ye~ ate     in t~ "\\b(e~ no                 neutral
6 ICE_GB~   958   958 know~ eat     it f~ "\\b(e~ yes                neutral
```

### Adding a case list

S. T. Gries (2013) recommends setting up the first column of the data frame such that it "numbers all $n$ cases from 1 to $n$ so that every row can be uniquely identified and so that you always restore one particular ordering (e.g., the original one)" (S. T. Gries 2013: 26). This is very easy to do: We specify a numeric vector ranging from 1 to the total number of rows in the data frame.

```
# Create a new Case column (which, by default, is added at the very end of
 ↪  the data frame)
kwic_data$Case <- 1:nrow(kwic_data)

# Move the Case column to the front of the data frame
```

```
kwic_data <- relocate(kwic_data, Case)

# Print reordered data frame
print(head(kwic_data))
```

```
# A tibble: 6 x 10
   Case docname        from    to pre   keyword post  pattern object_realisation
  <int> <chr>         <dbl> <dbl> <chr> <chr>   <chr> <chr>   <chr>
1     1 ICE_GB/S1A-0~   458   458 had ~ eaten   anyw~ "\\b(e~ no
2     2 ICE_GB/S1A-0~   478   478 : 1 ~ eating  will~ "\\b(e~ no
3     3 ICE_GB/S1A-0~   785   785 > Ye~ eat     befo~ "\\b(e~ no
4     4 ICE_GB/S1A-0~  1198  1198 the ~ eat     them~ "\\b(e~ yes
5     5 ICE_GB/S1A-0~  4529  4529 > Ye~ ate     in t~ "\\b(e~ no
6     6 ICE_GB/S1A-0~   958   958 know~ eat     it f~ "\\b(e~ yes
# i 1 more variable: verb_aspect <chr>
```

### Adding text metadata

Many research questions involve register, e.g., whether a construction is more frequent in speech than in writing. In the ICE corpora, information on the text category of each hit is stored in the `docname` column:

```
head(kwic_data$docname)
```

```
[1] "ICE_GB/S1A-006.txt" "ICE_GB/S1A-006.txt" "ICE_GB/S1A-006.txt"
[4] "ICE_GB/S1A-009.txt" "ICE_GB/S1A-009.txt" "ICE_GB/S1A-010.txt"
```

Each file name contains a **text category** label (e.g., `S1A`) followed by a **file number** (e.g., `001`). Depending on the version of the corpus files, the names may also contain a prefix (e.g., `ICE_GB/`) or the file extension `.txt`. Instead of splitting the names at fixed positions, it is therefore more robust to extract the relevant parts with regular expressions (see 2.2 Regular expressions):

```
kwic_data <- kwic_data %>%
  mutate(
    # S or W, followed by 1 or 2, followed by a letter from A to F
    text_category = str_extract(docname, "[SW][12][A-F]"),
    # three digits in a row
    file_number = str_extract(docname, "\\d{3}")
  )

kwic_data %>%
```

```
  select(docname, text_category, file_number) %>%
  head()
```

```
# A tibble: 6 x 3
  docname             text_category file_number
  <chr>               <chr>         <chr>
1 ICE_GB/S1A-006.txt S1A           006
2 ICE_GB/S1A-006.txt S1A           006
3 ICE_GB/S1A-006.txt S1A           006
4 ICE_GB/S1A-009.txt S1A           009
5 ICE_GB/S1A-009.txt S1A           009
6 ICE_GB/S1A-010.txt S1A           010
```

Summary of text categories in the ICE corpora

| Category | Subcategory | Type | Code Range |
|---|---|---|---|
| **SPOKEN (S)** | **DIALOGUE (S1)** | **PRIVATE (S1A)** | |
| | | Direct Conversations | S1A-001 to S1A-090 |
| | | Telephone Calls | S1A-091 to S1A-100 |
| | | **PUBLIC (S1B)** | |
| | | Class Lessons | S1B-001 to S1B-020 |
| | | Broadcast Discussions | S1B-021 to S1B-040 |
| | | Broadcast Interviews | S1B-041 to S1B-050 |
| | | Parliamentary Debates | S1B-051 to S1B-060 |
| | | Legal Cross-examinations | S1B-061 to S1B-070 |
| | | Business Transactions | S1B-071 to S1B-080 |
| | **MONOLOGUE (S2)** | **UNSCRIPTED (S2A)** | |
| | | Spontaneous Commentaries | S2A-001 to S2A-020 |
| | | Unscripted Speeches | S2A-021 to S2A-050 |

| | | | |
|---|---|---|---|
| | | Demonstrations | S2A-051 to S2A-060 |
| | | Legal Presentations | S2A-061 to S2A-070 |
| | | **SCRIPTED (S2B)** | |
| | | Broadcast News | S2B-001 to S2B-020 |
| | | Broadcast Talks | S2B-021 to S2B-040 |
| | | Non-broadcast Talks | S2B-041 to S2B-050 |
| **WRITTEN (W)** | **NON-PRINTED (W1)** | **NON-PROFESSIONAL WRITING (W1A)** | |
| | | Student Essays | W1A-001 to W1A-010 |
| | | Examination Scripts | W1A-011 to W1A-020 |
| | | **CORRESPONDENCE (W1B)** | |
| | | Social Letters | W1B-001 to W1B-015 |
| | | Business Letters | W1B-016 to W1B-030 |
| | **PRINTED (W2)** | **ACADEMIC WRITING (W2A)** | |
| | | Humanities | W2A-001 to W2A-010 |
| | | Social Sciences | W2A-011 to W2A-020 |
| | | Natural Sciences | W2A-021 to W2A-030 |
| | | Technology | W2A-031 to W2A-040 |
| | | **NON-ACADEMIC WRITING (W2B)** | |
| | | Humanities | W2B-001 to W2B-010 |

| | |
|---|---|
| Social Sciences | W2B-011 to W2B-020 |
| Natural Sciences | W2B-021 to W2B-030 |
| Technology | W2B-031 to W2B-040 |
| **REPORTAGE (W2C)** | |
| Press News Reports | W2C-001 to W2C-020 |
| **INSTRUCTIONAL WRITING (W2D)** | |
| Administrative Writing | W2D-001 to W2D-010 |
| Skills & Hobbies | W2D-011 to W2D-020 |
| **PERSUASIVE WRITING (W2E)** | |
| Press Editorials | W2E-001 to W2E-010 |
| **CREATIVE WRITING (W2F)** | |
| Novels & Stories | W2F-001 to W2F-020 |

If the fine-grained text categories are not needed, we can group them into macro-categories instead, such as **spoken** and **written** data. All spoken text categories begin with the letter `S`, all written ones with `W`.

```
kwic_data <- kwic_data %>%                                    ①
  mutate(medium = if_else(                                    ②
    str_starts(text_category, "S"),                           ③
    "spoken",                                                 ④
    "written"                                                 ⑤
  ))

kwic_data %>%
  select(text_category, medium) %>%
  head()
```

① Take the data frame `kwic_data` and pass it on to the next function via the pipe operator `%>%`. Store the result in `kwic_data` again.

② Create a new column with `mutate()` and call it `medium`. The column values are assigned conditionally with `if_else()`:
③ If `text_category` starts with the letter `S`, …
④ … classify `medium` as `"spoken"`, …
⑤ … otherwise as `"written"`.

```
# A tibble: 6 x 2
  text_category medium
  <chr>         <chr>
1 S1A           spoken
2 S1A           spoken
3 S1A           spoken
4 S1A           spoken
5 S1A           spoken
6 S1A           spoken
```

To examine only a subset of your data (e.g., spoken texts), use `filter()` to select the relevant rows:

```
kwic_data_spoken <- filter(kwic_data, medium == "spoken")

nrow(kwic_data_spoken)
```

```
[1] 71
```

## Where do I go from here?

As soon as you've fully annotated your dataset and reviewed it for potential coding errors, the next step involves analysing your data **statistically** to uncover potential patterns that are not visible to the naked eye. These patterns can include (subtle to major) differences in frequency of occurrence, relationships with other linguistically relevant features (e.g., register/genre) or probabilities (e.g., the probability that an object is omitted), among many others. Therefore, the following units will offer an introduction to statistical methods as they are typically applied in corpus linguistic research. A complete analysis of the *eat* data, from the research question to the write-up, is presented in 3.8 Case study.

## References

Goldberg, Adele E. 2001. "Patient Arguments of Causative Verbs Can Be Omitted: The Role of Information Structure in Argument Distribution." *Language Sciences* 23 (4): 503–24.

Gries, Stefan Thomas. 2013. *Statistics for Linguistics with r: A Practical Introduction*. 2nd rev. ed. Berlin; Boston: De Gruyter Mouton.

Gries, Stefan, and Sandra Deshors. 2014. "Using Regressions to Explore Deviations Between Interlanguage and Native Language." *Corpora* 9: 109–36.